

Import Data using Transform Maps (Spreadsheet)

ServiceNow Micro Project Documentation

Phase 2: Requirement Analysis

Item	Details
Project Title	Import Data using Transform Maps (Spreadsheet)
Programme	SkillWallet – Vetri Thiran Payirchi
Platform	ServiceNow
Team ID	SWTID-2026-2373

This phase captures what the Employee Data Import solution must do and the conditions it must operate under. The requirements are derived from the project workflow: preparing a spreadsheet, staging it in an Import Set table, transforming it into a custom target table, preventing duplicates with Coalesce, and reporting on the imported data.

2.1 Solution Requirements

2.1.1 Project Scope Summary

Item	Details
Project	Import Data using Transform Maps (Spreadsheet)
Platform	ServiceNow
Source	Sample Spreadsheet.xlsx (Google Sheets export), 15 employee records
Staging (Import Set) table	Employee Import [u_employee_import]
Target table	Employee Test [u_employee_test]
Transform Map	Sample Spreadsheet Import
Coalesce field	Employee ID
Outputs	3 reports and 1 dashboard (Employee Analytics Dashboard)

2.1.2 Functional Requirements

FR No.	Functional Requirement	Sub-Requirement / Description
FR-1	Source data preparation	Create a spreadsheet with columns Employee ID, Name, Email, Department and Location (15 sample records, SB-0001 to SB-0015) and download it as Sample Spreadsheet.xlsx.
FR-2	Custom target table	Create the table Employee Test (u_employee_test) with String fields Employee ID, Employee Name, Email, Department and Location, and confirm they appear on the form.

FR No.	Functional Requirement	Sub-Requirement / Description
FR-3	Load data into staging	Use Load Data to create the Import Set table Employee Import (u_employee_import) from the .xlsx file (Sheet number 1, Header row 1).
FR-4	Transform Map creation	Create the Transform Map Sample Spreadsheet Import from Employee Import to Employee Test and map all five source fields to the target fields using Auto Map Matching Fields or Mapping Assist.
FR-5	Transformation and validation	Run the transform and verify Completion code Success, then confirm all 15 records exist in the Employee Test table with a personalised list column order.
FR-6	Duplicate prevention (Coalesce)	Set Coalesce = true on the Employee ID field map. A repeat import must update existing records and insert only new Employee IDs.
FR-7	Import result verification	Use Transform History to check Total, Inserts, Updates, Ignored, Skipped and Errors after each import.
FR-8	Reporting	Create three reports on Employee Test: Employees by Department (Pie, Count), Employees by Location (Bar, Count) and Employee List Report (List).
FR-9	Dashboard	Create the Employee Analytics Dashboard, add all three reports to it, and restrict sharing by users, groups and roles where needed.
FR-10	Dashboard access control	Enable Admin overrides on the pa_dashboards create ACL so the dashboard form can be opened through PA_DASHBOARDS.FORM.

2.1.3 Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Steps use standard ServiceNow modules (Load Data, Transform Maps, Reports, Dashboards) so a beginner can follow them without scripting.
NFR-2	Data integrity	Field mapping must be complete and correct, and Coalesce on Employee ID must ensure no duplicate employee records exist.
NFR-3	Reliability	Each load and transform must finish with State Complete and Completion code Success, with errors shown as 0 in the import log.
NFR-4	Performance	The 15-record load processed in about 1.4 seconds and later 4-row imports ran in about one second, so bulk loads must complete promptly.
NFR-5	Security	Table creation, transform map changes and ACL changes require an administrator. Dashboard sharing can be limited by users, groups and roles.
NFR-6	Scalability	The staging plus transform approach must support repeat and larger imports without redesigning the table or map.
NFR-7	Maintainability	Clear names (u_employee_import, u_employee_test, Sample Spreadsheet Import) allow easy identification and later change of the configuration.

NFR No.	Non-Functional Requirement	Description
NFR-8	Data visibility	Reports and the dashboard must present department and location distribution and the full employee list from one view.

2.2 Data Requirements

2.2.1 Source Data Format

Column	Data Type	Example	Description
Employee ID	String	SB-0001	Unique employee identifier (coalesce key)
Name	String	Ratan Kumar	Employee full name
Email	String	test1@gmail.com	Employee email address
Department	String	ServiceNow	Department (ServiceNow, Salesforce)
Location	String	Hyderabad	Work location

2.2.2 Field Mapping (Data Dictionary)

Source Field (u_employee_import)	Target Field (u_employee_test)	Coalesce	Label
u_employee_id	u_employee_id	true	Employee ID
u_name	u_employee_name	false	Employee Name
u_email	u_email	false	Email
u_department	u_department	false	Department
u_location	u_location	false	Location

2.3 Expected Import Outcomes

Scenario	Input	Expected Result
Initial load	15 new employee records	15 inserts, 0 updates, 0 errors
Repeat load with changes (Coalesce on)	4 rows: 2 changed existing IDs, 2 new IDs	2 inserts, 2 updates, 0 errors (17 records total)
Repeat load with identical data	Same 4 rows again	0 inserts, 0 updates, 4 ignored

2.4 Software and Access Requirements

Requirement	Details
ServiceNow instance	Instance with access to Tables, Load Data, Transform Maps, Reports and Dashboards
User role	Administrator (needed for table creation and ACL update)
Spreadsheet tool	Google Sheets to prepare data; Excel (.xlsx) format for upload
Browser	Any modern web browser
Access control	pa_dashboards create ACL with Admin overrides enabled

2.5 Assumptions and Constraints

- The source file is a single-sheet .xlsx with the header in row 1.
- Employee ID is unique for each employee and is used as the Coalesce key.
- All target fields are String type, so no data type conversion is needed.
- Only Employee ID is coalesced. A change to any other field updates the record and does not create a new one.
- Dashboard creation through the backend name depends on the ACL change described in FR-10.

Please provide the next sub-topic or phase to continue.